

MLDM Assignment 4

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 4

This are your test results

 Recall

Derivatives 1

Status

Seen but not answered

Your score

0 / 3

0%

Response

Given $f(x) = 6x^3 - 9x + 4$, what is the derivative of $f(x)$?

☐

$f'(x) = 18x^2 - 3x + 9$

☐

$f'(x) = 18x^3 - 9x^2$

☐

$f'(x) = 18x^2 - 9$

☐

$f'(x) = 12x^2 - 7$

Derivatives 2

Status

Not started

Your score

0 / 1

0%

Response

Given $f(x) = -4x^5 - 5x^4 + 40x^3 + 6$, what is the derivative of $f(x)$?

☐

$f'(x) = -20x^2(x + 3)(x - 2)$

☐

$f'(x) = -x^4 + 20x^3 - 120x^2$

☐

$f'(x) = -20x^4 - 20x^3 + 120x^2$

☐

$f'(x) = 13x^2(2x + 3)(x + 2)$

Slope of curve 1

Status

Not started

Your score

0 / 5

0%

Response

Find the slope of the curve $(x + 2)^2 + 4$ at $x = 0$.

The slope of the given curve is:

⊙ Clustering evaluation

Status

Not started

Your score

0 / 1

0%

Response

Which of the following is not an evaluation metric for clustering

☐ Extrinsic

☐ Indirect

☐ Intrinsic

☐ Sensitivity

☑ Clustering types

Status

Not started

Your score

0 / 1

0%

Response

Which of the following are a clustering type?

☐ Hierarchical

☐ Regression

☐ Flat

☐ Aggregative

☰ Clustering types 2

Status

Not started

Your score

0 / 2.5

0%

Response

Choose right or wrong for each of the following statements

Notes:

• Correct answer: +0.5 pts

• Wrong answer: -0.25 pts

• Unanswered: 0 pts

Unanswered	Right	Wrong	
⊙	☐	☐	In flat clustering, there is no hierarchy and the number of clusters is chosen prior to clustering
⊙	☐	☐	With divisive clustering, to find the best split, we must explore all possibilities at each step.
⊙	☐	☐	Divisive clustering is a bottom-up approach where we start from first with the number of clusters equal to the number of data points. Then we continue pairing up smaller clusters into bigger ones.
⊙	☐	☐	Compared to hierarchical clustering, flat methods are much slower and computationally expensive.
⊙	☐	☐	Because there is no feedback loop in clustering, the model simply finds the output it deems best.

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3/8

⊙ Measures of proximity

Status

Not started

Your score

0 / 1

0%

Response

Which of the following is not a measure of proximity

☐ Cosine Similarity

☐ Manhattan Distance

☐ Transformation Function

☐ Simple Matching Coefficient

☐ Euclidian Distance

☐ Jaccard Coefficient

☐ Minkowski Distance

≡ Essay

Status

Not started

Your score

0 / 9.5

0%

Response

Explain in your own words the three main types of measurements how the quality of clustering can be evaluated **with examples**.

Note:

• Examples are required to get the full points.

0 word

 Practice

In the following questions, provide your answers with 4 digits after the decimal point.

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Dunn index

Status

Not started

Your score

0 / 5

0%

Response

Given the following clustering results (shown in the bellow figure). The objects have been inserted into three different clusters (C1, C2, and C3).

Scatter plot showing three clusters (C1, C2, and C3) on a 2D coordinate system. The x-axis ranges from 0 to 8, and the y-axis ranges from 0 to 4.5. The clusters are represented by shaded regions and labeled C1, C2, and C3. The points are labeled with their coordinates:

- C1 (Blue): (1, 2.5), (2.5, 2), (2.5, 3), (3.5, 3)
- C2 (Red): (4.5, 4), (5.5, 4.5), (5.5, 3.5), (6.5, 4)
- C3 (Green): (3.5, 1), (4.5, 1.5)

Compute the **Dunn Index** for the given clustering:

$DI_3 =$

Note:

- Use the **Euclidean distance**

Silhouette coefficient 1

Status

Not started

Your score

0 / 7

0%

Response

Given the following clustering results (shown in the bellow figure). The objects have been inserted into three different clusters (C1, C2, and C3).

Scatter plot showing three clusters (C1, C2, and C3) on a 2D coordinate system. The x-axis ranges from 0 to 8, and the y-axis ranges from 0 to 4.5. The clusters are represented by shaded regions and labeled C1, C2, and C3. The points are labeled with their coordinates:

- C1 (Blue): (1, 2.5), (2.5, 2), (2.5, 3), (3.5, 3)
- C2 (Red): (4.5, 4), (5.5, 4.5), (5.5, 3.5), (6.5, 4)
- C3 (Green): (3.5, 1), (4.5, 1.5)

Compute the **Silhouette coefficient** for the object 1: o1: (5.5, 3.5):

$s(o_1) =$

Note:

- Use the **Euclidean distance**

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Silhouette coefficient 2

Status

Not started

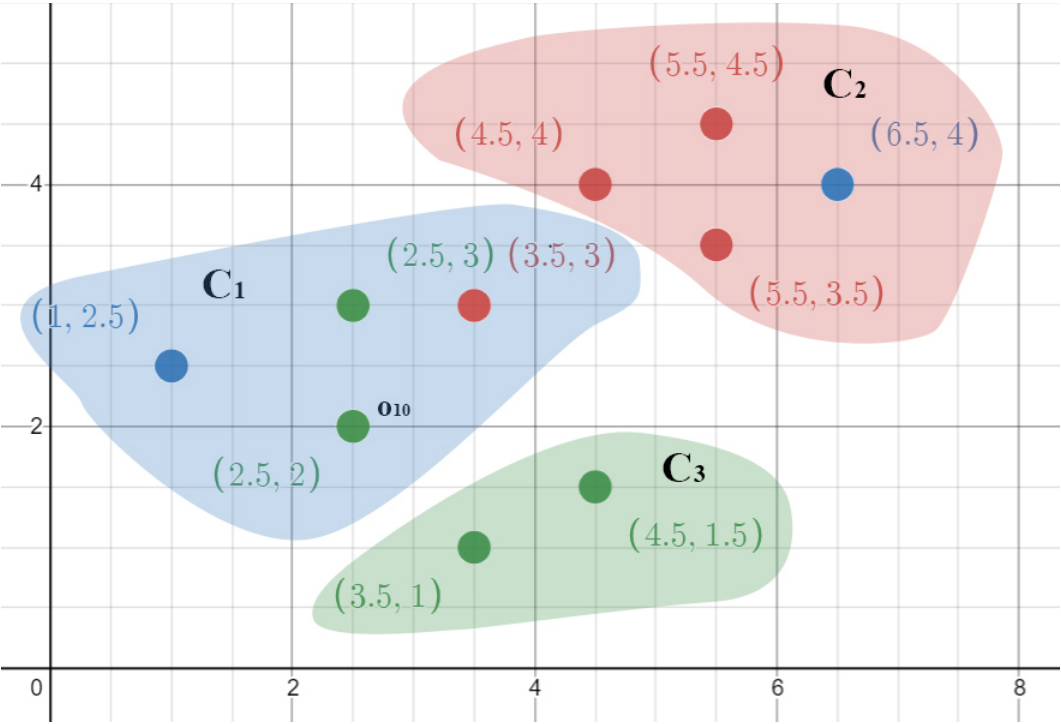
Your score

0 / 7

0%

Response

Given the following clustering results (shown in the bellow figure). The objects have been inserted into three different clusters (C1, C2, and C3).



Compute the **Silhouette coefficient** for the object 10:
o10: (2.5, 2):

$s(o_{10}) =$

Note:

- Use the **Euclidean distance**

Purity

Status

Not started

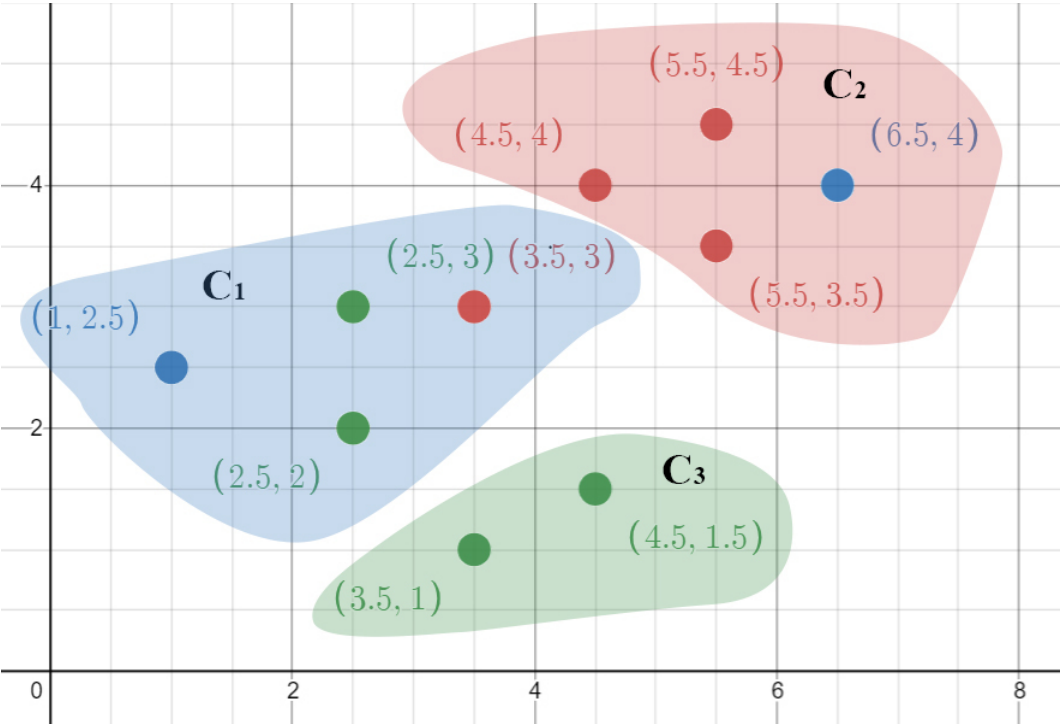
Your score

0 / 7

0%

Response

Given the following clustering results (shown in the bellow figure). The objects have been inserted into three different clusters (C1, C2, and C3).



Compute the **purity** over all clusters:

Purity =

 Programming

For this programming task, you are given a countries dataset containing the following attributes:

- name: the name of the country
- Longitude
- Latitude
- continent: the continent which if the country is located in

The dataset can be downloaded here: [Categorical.csv](#)

☒ Categorical features

Status Not started

Your score 0 / 3 0%

Response

We want to map the categorical column **"continent"** to an integer value representing each of the categories.

Example:

- Europe = 1
- Africa = 2
- Asia = 3
- ...etc

Which of the following is the correct line of code to achieve this task?

Note:

- The dataframe is provided in the "Programming" section's description (Categorical.csv)

- ☐ `dataframe.loc[:, 'continent'].map({'North America':0,'Europe':1,'Asia':2,'Africa':3,'South America':4, 'Oceania':5,'Seven seas (open ocean)':6, 'Antarctica':7})`
- ☐ `dataframe['continent'].map({'North America':0,'Europe':1,'Asia':2,'Africa':3,'South America':4, 'Oceania':5,'Seven seas (open ocean)':6, 'Antarctica':7})`
- ☐ `dataframe.iloc[:, 'continent'].map({'North America':0,'Europe':1,'Asia':2,'Africa':3,'South America':4, 'Oceania':5,'Seven seas (open ocean)':6, 'Antarctica':7})`
- ☐ `dataframe.iloc[:, 3].map({'North America':0,'Europe':1,'Asia':2,'Africa':3,'South America':4, 'Oceania':5,'Seven seas (open ocean)':6, 'Antarctica':7})`
- ☐ `dataframe.loc['continent'].map({'North America':0,'Europe':1,'Asia':2,'Africa':3,'South America':4, 'Oceania':5,'Seven seas (open ocean)':6, 'Antarctica':7})`
- ☐ `dataframe['continent'].apply({'North America':0,'Europe':1,'Asia':2,'Africa':3,'South America':4, 'Oceania':5,'Seven seas (open ocean)':6, 'Antarctica':7})`

 Feature selection

Status Not started

Your score 0 / 5 0%

Response

We would like to select only the column "continent" as a feature for our model.

Which of the following lines of code is correct to select only the "continent" feature?

Unanswered Right Wrong

- ☐ ☐ ☐ `x = dataframe.iloc[:, "continent"]`
- ☐ ☐ ☐ `x = dataframe[:,3:4]`
- ☐ ☐ ☐ `x = dataframe.iloc[:, 3:4]`
- ☐ ☐ ☐ `x = dataframe.iloc[:, 4]`
- ☐ ☐ ☐ `x = dataframe.loc[:, "continent"]`

 k-means clustering

Status

Not started

Your score

0 / 15

0%

Response

Write a K-Means algorithm using python code to cluster our data (Categorical.csv) into 4 different clusters using only the column "continent" as a feature. Finally, visualize your data (countries) using longitude and latitude in a scatter plot (**the data points need to have a color indicating which cluster they belong to**).

The colors of the clusters needs to be:

- Cluster 1: Red
- Cluster 2: Green
- Cluster 3: Blue
- Cluster 4: Black

Please upload the visualization plot achieved by your code in a form of JPG or PNG.

Notes:

- The plot needs to have x and y labels + a title for the plot.
- The image needs to be visible
- Libraries allowed are pandas, NumPy, and matplotlib.

File

No file uploaded

 Upload file

Status

Not started

Your score

0 / 15

0%

Response

Please upload the **jupyter notebook** used to solve the task "**K-Means Clustering**".

File

No file uploaded

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